

Day 2 — Networking Basics

Elaborated Study Notes: IP Addresses · DNS · Ports & Protocols · Practical Commands

1. Learning Objectives

- Explain what a computer network is and identify everyday examples.
- Understand what IP addresses are and why devices need them.
- Understand how DNS translates human-friendly domain names into IP addresses.
- Understand ports and common protocols such as HTTP, HTTPS, SSH and FTP.
- Use basic commands such as ping, nslookup and tracert/traceroute for network diagnostics.

2. What Is a Network?

Definition: A network is a group of computers and other devices connected together so they can share data, resources and communicate. Connections may be wired, such as Ethernet, or wireless, such as Wi-Fi. A network can be as small as a room or as large as the global Internet.

Everyday examples

- **Internet:** A global network connecting billions of devices worldwide.
- **Home Wi-Fi:** Phones, laptops, smart TVs and other devices communicate through a router.
- **College/office LAN:** Computers and other systems within a building can communicate over a Local Area Network (LAN).

Why networking matters: Without networks, devices would be isolated. Browsing websites, sending email, using messaging applications and accessing online services all depend on devices being able to locate and communicate with other systems.

Key idea: A network provides the communication path, while addressing and protocols help devices know where and how to communicate.

3. IP Addresses

Definition: An IP (Internet Protocol) address is a numerical address assigned to a device on a network. It helps identify where data should be delivered, much like a postal address helps mail reach the correct house.

Example: 192.168.1.1

Types of IP addresses

Type	Explanation	Example
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IPv4	32-bit address written as four numbers separated by dots. Each number normally ranges from 0 to 255.	192.168.1.1
IPv6	128-bit address created to support a much larger number of connected devices.	2001:0db8::1
Private IP	Used inside local networks such as homes and offices and not directly reachable from the public Internet.	10.0.0.1 or 192.168.x.x
Public IP	Assigned by an Internet Service Provider and used to identify a device on the Internet.	ISPs assign them

Important distinction: Private IP addresses are used within local networks, while public IP addresses are used for communication across the Internet.

4. DNS — Domain Name System

Definition: DNS is the system that converts human-friendly website names into machine-readable IP addresses. People remember names such as **google.com**, while computers communicate using IP addresses. DNS connects these two forms of identification.

Example: google.com → 142.250.x.x

Think of DNS as the Internet's phonebook. When you enter a domain name into a browser, DNS looks up the matching IP address so your device knows where to send the request.

5. Ports and Protocols

Port: A port is a numbered communication endpoint on a device. It allows multiple services and applications to operate on the same device without their data becoming mixed together. The IP address identifies the device, while the port identifies the specific service.

Common ports and protocols

Port	Protocol	Typical use
80	HTTP	Regular, unencrypted website traffic
443	HTTPS	Secure, encrypted website traffic
22	SSH	Secure remote login to a server
21	FTP	Transferring files between computers

Protocol: A protocol is a set of rules that defines how devices communicate. HTTP and HTTPS are used for web communication, SSH provides secure remote access, and FTP is used for file transfer.

6. How the Internet Works — Simple Website Flow

The networking concepts become easier to understand when combined into one practical sequence showing what happens after a user enters a website address.

- 1 The user types google.com into the browser.

- 2 The browser asks DNS to find the IP address linked to that domain name.
- 3 DNS returns the matching IP address, for example 142.250.x.x.
- 4 The browser sends a request to the server using that IP address on port 443 for HTTPS.
- 5 The server processes the request and sends back a response.
- 6 The browser receives the response and renders the website on the screen.

Putting it together

Domain name → DNS → IP address → Port/Service → Server response → Browser displays the webpage.

7. Practical Network Commands

7.1 ping — Check Connectivity

The ping command sends small test packets to a target and checks whether it responds. It also measures the round-trip time, making it useful for basic connectivity testing.

Command:

```
ping google.com
```

Observe whether replies are received and the response time shown by the command.

7.2 nslookup — Find IP Address

The nslookup command queries DNS directly and displays the IP address associated with a domain name. It is useful for checking which address DNS returns for a website.

Command:

```
nslookup google.com
```

7.3 tracert / traceroute — Trace the Route

These commands display the intermediate routers, called hops, that data travels through on the way to a destination. They are useful for identifying where a network connection may become slow or fail.

Windows:

```
tracert google.com
```

Linux / macOS:

```
traceroute google.com
```

8. Task

The following IP addresses are examples from the original notes. Exact results may vary depending on location, ISP and the website's load-balanced infrastructure.

Website	Example IPv4 address
google.com	142.250.179.206
github.com	140.82.112.3
wikipedia.org	208.80.154.224

DNS in simple words: DNS is like the Internet's phonebook. People remember website names, while computers communicate using numeric IP addresses. DNS finds the matching IP address and helps the browser connect to the correct server.

Three common ports and their use: 80 — HTTP for regular web traffic, 443 — HTTPS for secure web traffic, and 22 — SSH for secure remote login.